Predictive Modeling and Feature Selection: An Analysis of the Ames Housing Dataset

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Introduction & Data Preprocessing

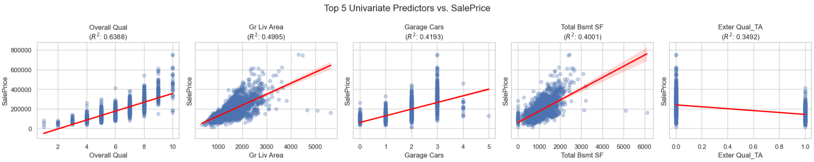
Objective: Build optimized prognostic (pricing) and diagnostic (classification) models using the Ames Housing dataset.

Data Cleaning & Stability Metrics:

- **Initial State:** 2930 observations, 82 explanatory variables(39 are Quantitative and 43 are Qualitative).
- **Dummy Encoding:** Handled categorical variables, expanding the matrix to 257 features.
- **Multicollinearity Resolution:** Dropped perfectly collinear variables (e.g., `Total Bsmt SF`, `Garage Qual_None`) to resolve VIF = ∞.
- **Outlier Removal:** Removed 195 influential points (Cook's D $> 4/n$) and 44 extreme outliers ($|\text{Std. Resid}| > 3$).

Univariate Analysis: Top Predictors

Methodology: Fitted 257 Simple Linear Regression models to isolate the individual predictive power of each feature.



Univariate Analysis: Interpretations

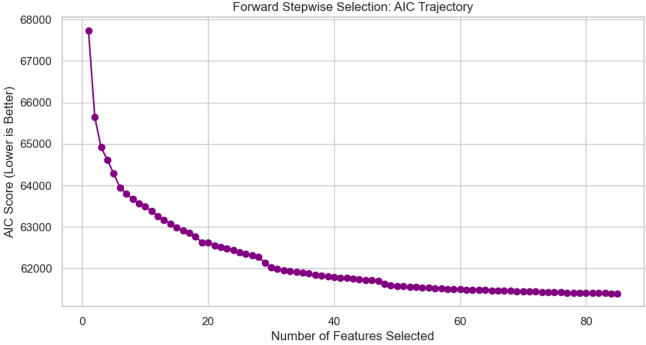
Top 5 Individual Predictors (R^2):

1. **Overall Qual:** 0.6388
2. **Gr Liv Area:** 0.4995
3. **Garage Cars:** 0.4193
4. **Total Bsmt SF:** 0.4001
5. **Exter Qual_TA:** 0.3492

Interpretation: Quality and scale (Square Footage) drive the market. `Overall Qual` alone explains almost 64% of price variance, validating its heavy weighting in subsequent multivariate models.

Feature Selection: Forward Stepwise

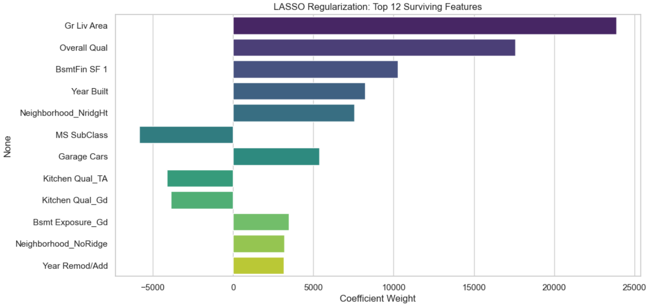
Goal: Compare subset selection strategies to bypass the curse of dimensionality.



Result: Stepwise selection greedily drafted 85 columns. Despite a high Adjusted R^2 (0.9406), the AIC (61,217) indicates severe overfitting and unnecessary model complexity.

Feature Selection: LASSO Shrinkage

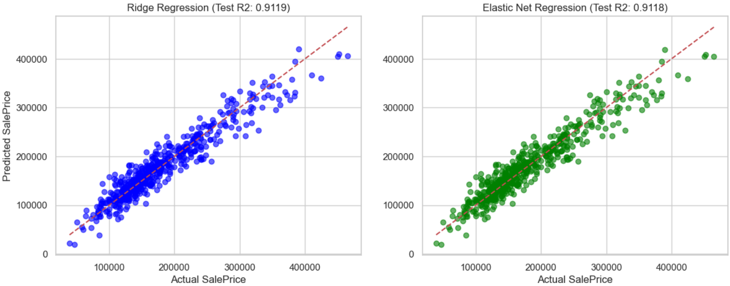
Goal: Apply an L_1 penalty to shrink redundant noise and find the optimal subset.



Result: LASSO ($\alpha = 1200$) isolated exactly **12 features**. The model achieved an AIC of 50,810. The lower AIC proves LASSO achieved a vastly superior Bias-Variance tradeoff compared to Stepwise selection.

Regularization Validation: Ridge & Elastic Net

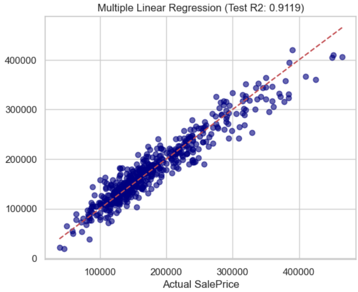
Validation: We applied L_2 and Hybrid penalties to the final 12 LASSO features to stress-test for residual multicollinearity.

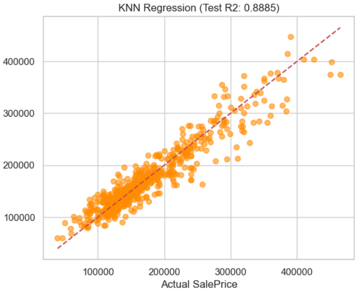


Interpretation: Ridge ($R^2 = 0.8948$) and Elastic Net ($R^2 = 0.8955$) performed nearly identically to unpenalized OLS. Elastic Net automatically selected an L_1 ratio of 1.0, mathematically proving that pure LASSO was the optimal feature selector.

Prognostic Modeling: OLS vs. KNN

Methodology: Predicting continuous `SalePrice` using the 12 optimal features.

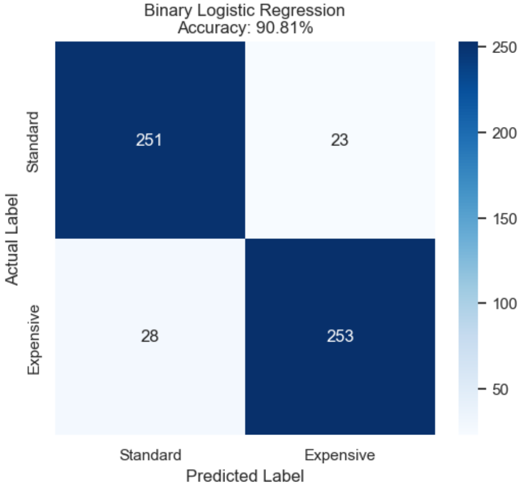




Interpretation: Multiple Linear Regression ($R^2 = 0.8958$) slightly outperformed KNN Regression ($R^2 = 0.8878$). Housing pricing metrics scale highly linearly, making standard OLS coefficient weighting highly effective and robust.

Diagnostic Modeling: Binary Classification

Methodology: Engineered a binary target (”Standard” vs. ”Expensive” based on median price) using Logistic Regression on the 12 selected features.



Diagnostic Modeling: Multi-Category Classification

Methodology: Categorized the market into three tiers (Budget, Standard, Luxury).

Conclusion

Final Project Takeaways:

- **Dimensionality Reduction:** Shrinkage methods (LASSO) successfully reduced 257 dimensions to 12 ($\sim$5%), entirely eliminating the noise captured by Forward Stepwise selection.
- **Predictive Power:** The sparse 12-feature matrix explained $\sim$90% of the continuous pricing variance (RMSE $\approx$ $22,600$).
- **Model Versatility:** The selected features proved mathematically robust across both prognostic algorithms (OLS/KNN) and diagnostic classification algorithms (Logistic), confirming the underlying stability of the Ames real estate market signals.